ASSESSMENT TOOL 3

DEBUGGING & OPTIMISATION TASK

CSA0609-DESIGN AND ANALYSIS OF ALGORITHMS

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Submitted by

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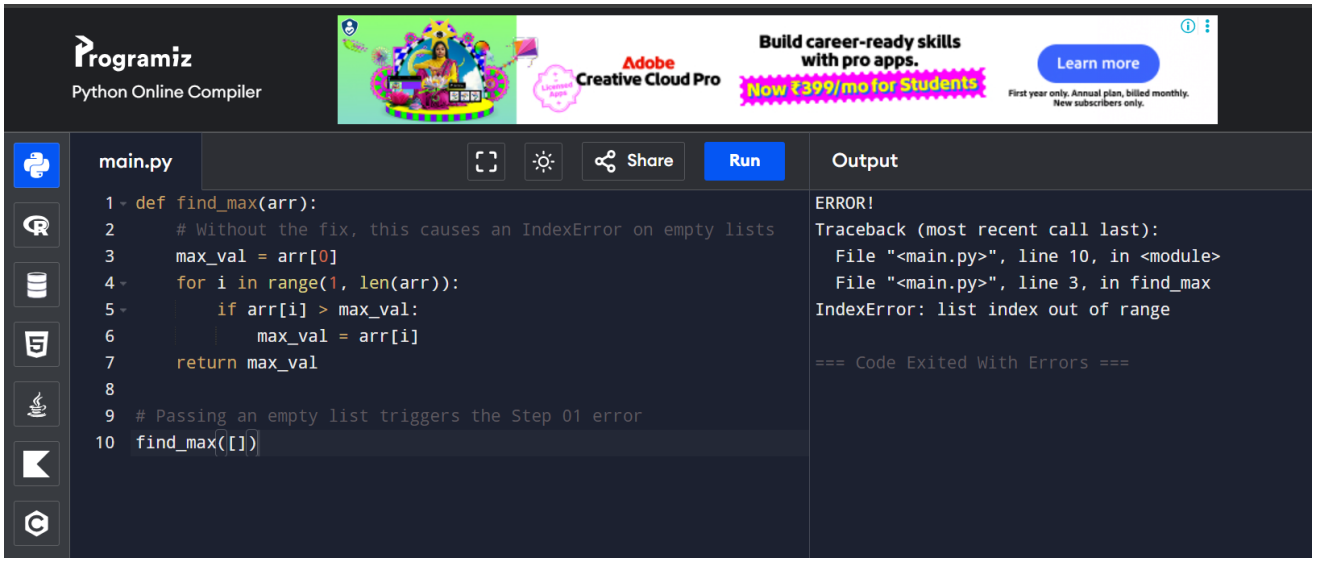
Debugging Maximum Element Search Code (Empty List and Initialization Risk)

The following code is intended to find the maximum element, but it fails for empty arrays and uses an unsafe initialization pattern that may lead to errors. Carefully analyze the assumptions made by the code and explain why robustness is affected. Apply systematic debugging to handle edge cases correctly while preserving proper maximum search behavior. Optimize the implementation for clarity and defensive programming. Analyze the time complexity of the corrected solution and rewrite the final code with suitable documentation.

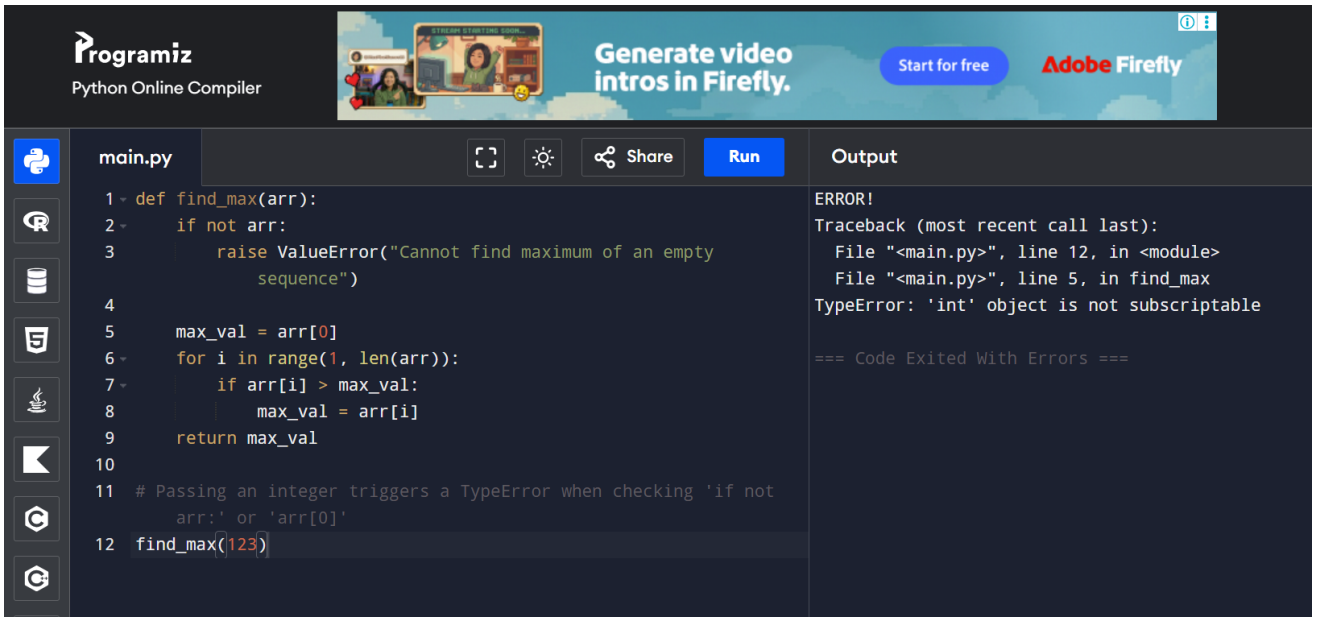
```
def find_max(arr):
    max_val = arr[0]   # ISSUE
    for i in range(1, len(arr)):
        if arr[i] > max_val:
            max_val = arr[i]
    return max_val
```

STEP BY STEP ERROR ANALYSIS :

STEP 01 : Direct indexing of arr[0] without checking if arr is non-empty.



STEP 02 : Strict assumption that the input arr is always a valid indexable sequence.



EFFECTS OF ERROR :

1. If an empty list [] is passed into find_max([]), Python raises an IndexError: list index out of range, causing the program to crash immediately.
2. If None or a non-sequence data type is passed into the function, it causes an unhandled TypeError prior to or during length evaluation.

CORRECT STEPS :

1. Add defensive checks to handle empty input before accessing indices, such as returning None, raising a descriptive ValueError, or using a default initial value.
2. Perform type validation or ensure structural input checks so the code handles invalid input cleanly.

ALGORITHM

1. **Input:** Accept a collection arr.
2. **Empty / Validity Check:** Check if arr is None or empty (len(arr) == 0).
    - If empty or invalid, raise a ValueError("Cannot find maximum element of an empty sequence") or return None (depending on application requirements; raising ValueError mirrors standard Python max() behavior).
3. **Initialization:** Set max_val to the first element arr[0].
4. **Iteration:** Loop through each element item in arr[1:] (or iterate directly across elements).
5. **Comparison:** If item is strictly greater than max_val, update max_val = item.
6. **Return Result**: Return max_val.

MANUAL PROCESS

Scenario A: Normal Execution (arr = [3, 7, 2, 9, 5])

1. Initial check: arr is not empty. Proceed.
2. Initialize: max_val = arr[0] → max_val = 3.
3. Loop $i = 1$: arr[1] = 7. Compare: 7 > 3 is True. Update max_val = 7.
4. Loop $i = 2$: arr[2] = 2. Compare: 2 > 7 is False. Keep max_val = 7.
5. Loop $i = 3$: arr[3] = 9. Compare: 9 > 7 is True. Update max_val = 9.
6. Loop $i = 4$: arr[4] = 5. Compare: 5 > 9 is False. Keep max_val = 9.
7. End of Loop: Return 9.

Scenario B: Edge Case (arr = [])

1. **Initial check:** arr is empty (len([]) == 0).
2. **Defensive trigger:** Instead of attempting to evaluate arr[0] (which would crash), the defensive check instantly raises ValueError("Cannot find maximum of an empty sequence") (or returns None).
3. **Outcome:** The program handles the boundary condition gracefully without unhandled crashes.

COMPLEXITIES

- **Time Complexity: $\mathcal{O}(n)$**
  - ***Explanation**:* The function visits every element of the array of size n exactly once during iteration to find the maximum value.
- **Space Complexity: $\mathcal{O}(1)$**
  - ***Explanation:*** The memory footprint remains constant because only a single auxiliary variable (max_val) is used, regardless of the size of arr.

DEBUGGING SUMMARY :

- Primary Flaw: The original function relies on direct index access (arr[0]) without verifying whether the input sequence contains any elements.
- Failure Scenario: When an empty list ([]) is passed, evaluating arr[0] raises an uncaught IndexError, making the code fragile and prone to sudden runtime crashes.
- Risks of Arbitrary Initializers: Initializing max_val with arbitrary external values (like 0 or float('-inf')) introduces subtle logical bugs, particularly when dealing with lists containing negative numbers or custom objects.
- Defensive Programming Solution: The implementation must validate that the input array arr is truthy and non-empty before attempting to access any indices.
- Explicit Fault Signaling: Raising an explicit ValueError on empty inputs—following the standard behavior of Python's built-in max() function—or returning None prevents unexpected crashes.
- Overall Outcome: Enforcing proper input validation guarantees robust, error-free max-search execution across all boundary edge cases.

PSEUDO CODE :

```
FUNCTION find_max(arr):
   // Check if the sequence is empty or invalid
   IF arr IS EMPTY OR arr IS NULL THEN
      RAISE ValueError("Cannot find maximum of an empty sequence")
   END IF

   // Initialize the max tracker with the first element
   SET max_val TO arr[0]

   // Iterate through the remaining elements
   FOR EACH element FROM index 1 TO length(arr) - 1 DO
      IF arr[index] > max_val THEN
         SET max_val TO arr[index]
      END IF
   END FOR

   RETURN max_val
  END FUNCTION
```

CORRECTED & OPTIMIZED IMPLEMENTATION

CODE :

```
def find_max(arr):
    # Step 01 Fix: Check if array is empty before accessing arr[0]
    if not arr:
        raise ValueError("Cannot find maximum of an empty sequence")

    max_val = arr[0]
    for i in range(1, len(arr)):
        if arr[i] > max_val:
            max_val = arr[i]
    return max_val
```

DRIVER / TEST CODE

```
# 1. Normal list of integers
nums1 = [3, 7, 2, 9, 5]
print("Input :", nums1)
print("Output:", find_max(nums1))
print("-" * 30)

# 2. List with negative numbers
nums2 = [-10, -3, -50, -1]
print("Input :", nums2)
print("Output:", find_max(nums2))
print("-" * 30)

# 3. Single-element list
nums3 = [42]
print("Input :", nums3)
print("Output:", find_max(nums3))
print("-" * 30)

# 4. Testing the empty list edge case (raises ValueError)
try:
    print("Input : []")
    print("Output:", find_max([]))
except ValueError as e:
    print("Output: Caught Error ->", e)
```